

Harvard's Endowment: Mean-Variance Analysis

FINM 36700 Portfolio Theory and Risk Management

Ivy Chan | Jennifer Xiao | Edouard Lavalard | Alexei Le

2. Mean-Variance Optimization

2.1 Summary statistics

SPY has the highest Sharpe ratio (0.90), almost twice that of the next asset, HYG (0.47). BWX has the lowest (−0.21): it is the only asset with a negative mean excess return, as foreign bonds lost to a strong dollar over the sample. DBC is second lowest (0.08). Table 1 reports the full set.

Table 1: Annualized excess-return statistics, sorted by Sharpe ratio

Asset	Class	Mean	Volatility	Sharpe
SPY	Domestic equity	12.70%	14.14%	0.90
HYG	High yield	3.45%	7.33%	0.47
EFA	Foreign equity	6.38%	14.86%	0.43
IYR	Real estate	7.01%	16.55%	0.42
PSP	Private equity	7.64%	21.13%	0.36
TIP	Inflation-indexed	1.28%	4.97%	0.26
EEM	Emerging markets	4.44%	17.91%	0.25
IEF	Domestic bonds	0.86%	6.21%	0.14
DBC	Commodities	1.30%	17.23%	0.08
BWX	Foreign bonds	−1.72%	8.19%	−0.21

2.2 Descriptive analysis

The highest correlation is between PSP and SPY (0.87): listed private equity behaves like levered US equity. The lowest is between DBC and IEF (−0.31): commodities rally on inflation shocks, which hurt nominal Treasuries. Figure 1 shows the full matrix.

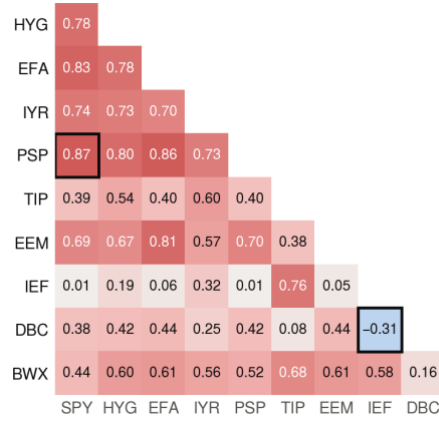


Figure 1: Correlation of monthly excess returns. Assets are ordered by Sharpe ratio; the highest and lowest pairs are outlined.

TIPS outperformed both domestic and foreign bonds over the sample, with a higher mean, a lower volatility and a higher Sharpe ratio (0.26, against 0.14 for IEF and −0.21 for BWX). In absolute terms the performance remains modest, at 1.3% a year over cash (Table 2).

Table 2: TIPS against domestic and foreign bonds (annualized)

Asset	Mean	Volatility	Sharpe
TIP (Inflation-indexed)	1.28%	4.97%	0.26
IEF (Domestic bonds)	0.86%	6.21%	0.14
BWX (Foreign bonds)	−1.72%	8.19%	−0.21

2.3 The MV frontier

The tangency portfolio is computed from the sample mean $\tilde{\mu}$ and covariance Σ of monthly excess returns:

$$w^{tan} = \frac{\Sigma^{-1} \tilde{\mu}}{\mathbf{1}' \Sigma^{-1} \tilde{\mu}}.$$

The ranking of the weights does not follow the ranking of the Sharpe ratios (Figure 2). IEF receives the largest weight (1.32) despite a Sharpe ratio of only 0.14: its correlation with equities is close to zero and its correlation with DBC is negative, so it hedges the equity exposure. Conversely, TIP, IYR and PSP are shorted although their Sharpe ratios are positive, because they are largely spanned by assets already held: IEF for TIP (correlation 0.76) and SPY for IYR and PSP (0.74 and 0.87). The weights reflect each asset's Sharpe ratio after hedging with the other assets, not its standalone Sharpe ratio.

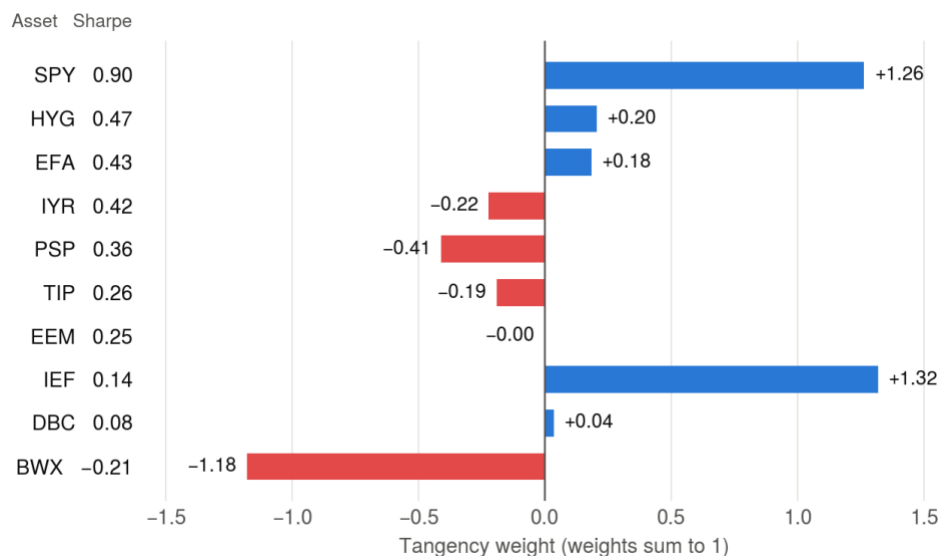


Figure 2: Tangency weights, with assets ordered by standalone Sharpe ratio (shown next to each ticker). Long positions in blue, short positions in red.

The tangency portfolio has an annualized mean of 16.17%, a volatility of 10.96% and a Sharpe ratio of 1.48.

2.4 TIPS

Table 3 reports the tangency portfolio in three cases: the base case, without TIPS, and with the expected excess return of TIPS raised by 0.0012 per month (1.44% a year, from 1.28% to 2.72%).

Table 3: Tangency weights and performance under the TIPS scenarios

Asset	Base	No TIPS	TIPS mean +0.0012
IEF	1.32	1.16	0.13
SPY	1.26	1.21	0.88
HYG	0.20	0.18	0.02
EFA	0.18	0.19	0.21
DBC	0.04	0.03	-0.04
EEM	-0.00	-0.01	-0.04
TIP	-0.19		1.26
IYR	-0.22	-0.22	-0.22
PSP	-0.41	-0.40	-0.33
BWX	-1.18	-1.14	-0.86
Mean	16.17%	15.54%	13.27%
Volatility	10.96%	10.54%	8.42%
Sharpe	1.476	1.475	1.575

Dropping TIPS barely changes the result: the Sharpe ratio moves from 1.476 to 1.475 and the other weights hardly change, so TIPS are almost spanned by IEF, BWX and the other assets in this sample. Raising their mean by 0.0012 per month, however, takes the TIP weight from -0.19 to $+1.26$, at the expense of IEF (1.32 to 0.13) and HYG (0.20 to 0.02), and the Sharpe ratio rises to 1.57. This adjustment is about one standard error of the sample mean of TIP (1.26% a year), so it lies well within estimation noise.

On the historical data, TIPS therefore do not expand the investment opportunity set. The case for treating them as a separate asset class has to rest on forward-looking assumptions, such as the 4% real yield used by HMC, and on Harvard's real-return objective, which this nominal analysis does not capture. Harvard is right to consider TIPS separately, but the sample statistics alone do not justify it.

3. Allocations

Each allocation is rescaled to the target monthly excess mean $\tilde{\mu}^p = 0.01$:

$$w = \tilde{\mu}^p \frac{v}{v' \tilde{\mu}}, \quad v_i^{EW} = 1, \quad v_i^{RP} = \frac{1}{\sigma_i^2}, \quad v^{MV} = \Sigma^{-1} \tilde{\mu}.$$

Table 4: Allocation weights at the 1% monthly target

Asset	EW	RP	MV
IEF	0.28	1.17	0.98
SPY	0.28	0.23	0.94
HYG	0.28	0.84	0.15
EFA	0.28	0.20	0.14
DBC	0.28	0.15	0.03
EEM	0.28	0.14	-0.00
TIP	0.28	1.82	-0.14
IYR	0.28	0.16	-0.17
PSP	0.28	0.10	-0.30
BWX	0.28	0.67	-0.88
Sum of weights	2.77	5.50	0.74

Table 5: Full-sample performance of the allocations (annualized)

Portfolio	Mean	Volatility	Sharpe
EW	12.00%	27.53%	0.44
RP	12.00%	32.61%	0.37
MV	12.00%	8.13%	1.48

All three portfolios earn 12% a year by construction, so they differ only in risk (Table 5). The mean-variance portfolio reaches the target with 8.1% volatility and a Sharpe ratio of 1.48, against 27.5% (0.44) for equal-weight and 32.6% (0.37) for risk-parity.

- Mean-variance dominates by construction: in sample, the tangency portfolio is the maximum-Sharpe portfolio. It reaches the target with a net exposure below one (weights sum to 0.74)
- Equal-weight has aggregate risky-asset weights of 2.77 and is effectively an equity-heavy long-only portfolio
- Risk-parity performs worst. Inverse-variance weighting concentrates capital in low-volatility bonds (TIP 1.82, IEF 1.17, HYG 0.84, BWX 0.67), which have low or negative Sharpe ratios in this sample, and reaching 12% then requires aggregate risky-asset weights of 5.50

The advantage of mean-variance is an upper bound, since the portfolio is evaluated on the same means and covariances it was estimated from.

Appendix: Code

```
import pandas as pd, numpy as np
X = pd.read_excel('multi_asset_etf_data.xlsx', sheet_name='excess returns', index_col=0)
R = X.drop(columns='QAI')
MU_TGT = 0.01
def ann(r):
    r = r.to_frame() if isinstance(r, pd.Series) else r
    m, v = r.mean() * 12, r.std() * np.sqrt(12)
    return pd.DataFrame({'mean': m, 'vol': v, 'sharpe': m / v})
def tangency(r, mu=None):
    mu = r.mean() if mu is None else mu
    w = np.linalg.solve(r.cov(), mu)
    return pd.Series(w / w.sum(), r.columns)
def port_stats(w, r, mu=None):
    mu = r.mean() if mu is None else mu
    m, v = w @ mu * 12, np.sqrt(w @ r.cov() @ w * 12)
    return pd.Series({'mean': m, 'vol': v, 'sharpe': m / v})
stats = ann(R).sort_values('sharpe', ascending=False)
corr = R.corr()
w_tan = tangency(R)
w_noTIP = tangency(R.drop(columns='TIP'))
mu_adj = R.mean().copy(); mu_adj['TIP'] += 0.0012
w_adj = tangency(R, mu_adj)
def allocations(r, tgt=MU_TGT):
    mu = r.mean()
    raw = {'EW': pd.Series(1., r.columns), 'RP': 1 / r.var(), 'MV': tangency(r)}
    return pd.DataFrame({k: w * tgt / (w @ mu) for k, w in raw.items()})
W = allocations(R)
perf = pd.DataFrame({k: ann(R @ W[k]).iloc[0] for k in W}).T
```